PREMIUM

Full-Length Mock Test

Mock 02

Computer Science and Information Technology

GATE CS

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Algorithms — Divide & Conquer]

Which data structure implements FIFO?

A) Stack
B) Queue
C) Tree
D) Graph

Q.2 [1 Mark] [EASY] [Data Structures — Hashing]

Time complexity of binary search?

A) O(n)
B) O(log n)
C) O(1)
D) $O(n^2)$

Q.3 [1 Mark] [EASY] [DBMS — Indexing]

Normal form where all non-prime attributes depend on key:

A) 1NF
B) 2NF
C) 3NF
D) BCNF

Q.4 [1 Mark] [EASY] [OS — Deadlock]

In deadlock prevention, hold-and-wait is prevented by:

A) Preemption
B) Resource ordering
C) Mutual exclusion
D) Allocation at once

Q.5 [1 Mark] [EASY] [CN — Routing]

Class A IP range:

A) 0-127
B) 128-191
C) 192-223
D) 224-239

Q.6 [1 Mark] [EASY] [TOC — Decidability]

A regular grammar generates:

A) Type 0
B) Type 1
C) Type 2
D) Type 3

Q.7 [1 Mark] [EASY] [COA — Pipelining]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

A) 0.1

B) 0.5

C) 0.9

D) 0.99

Q.8 [1 Mark] [EASY] [Digital Logic — Combinational Circuits]

Which is not a page replacement algorithm?

A) FIFO
B) LRU
C) Optimal
D) FCFS

Q.9 [1 Mark] [EASY] [Engineering Mathematics — Discrete Math]

Which data structure implements FIFO?

A) Stack
B) Queue
C) Tree
D) Graph

Q.10 [1 Mark] [EASY] [Programming — C Programming]

Time complexity of binary search?

A) O(n)
B) O(log n)
C) O(1)
D) $O(n^2)$

Q.11 [1 Mark] [EASY] [Algorithms — Greedy]

Normal form where all non-prime attributes depend on key:

A) 1NF
B) 2NF
C) 3NF
D) BCNF

Q.12 [1 Mark] [EASY] [Data Structures — Linked Lists]

In deadlock prevention, hold-and-wait is prevented by:

A) Preemption
B) Resource ordering
C) Mutual exclusion
D) Allocation at once

Q.13 [2 Marks] [MODERATE] [DBMS — Concurrency]

Class A IP range:

A) 0-127
B) 128-191
C) 192-223
D) 224-239

Q.14 [2 Marks] [MODERATE] [OS — Process Sync]

A regular grammar generates:

A) Type 0

B) Type 1

C) Type 2

D) Type 3

Q.15 [2 Marks] [MODERATE] [CN — TCP/IP]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

A) 0.1
B) 0.5
C) 0.9
D) 0.99

Q.16 [2 Marks] [MODERATE] [TOC — Grammars]

Which is not a page replacement algorithm?

A) FIFO
B) LRU
C) Optimal
D) FCFS

Q.17 [2 Marks] [MODERATE] [COA — Addressing Modes]

Which data structure implements FIFO?

A) Stack
B) Queue
C) Tree
D) Graph

Q.18 [2 Marks] [MODERATE] [Digital Logic — Minimization]

Time complexity of binary search?

A) O(n)
B) O(log n)
C) O(1)
D) $O(n^2)$

Q.19 [2 Marks] [MODERATE] [Engineering Mathematics — Probability]

Normal form where all non-prime attributes depend on key:

A) 1NF
B) 2NF
C) 3NF
D) BCNF

Q.20 [2 Marks] [MODERATE] [Programming — Recursion]

In deadlock prevention, hold-and-wait is prevented by:

A) Preemption
B) Resource ordering
C) Mutual exclusion
D) Allocation at once

Q.21 [2 Marks] [MODERATE] [Algorithms — Dynamic Programming]

Class A IP range:

A) 0-127

B) 128-191

C) 192-223

D) 224-239

Q.22 [2 Marks] [MODERATE] [Data Structures — Graphs]

A regular grammar generates:

A) Type 0
B) Type 1
C) Type 2
D) Type 3

Q.23 [2 Marks] [MODERATE] [DBMS — Normalization]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

A) 0.1
B) 0.5
C) 0.9
D) 0.99

Q.24 [2 Marks] [MODERATE] [OS — Deadlock]

Which is not a page replacement algorithm?

A) FIFO
B) LRU
C) Optimal
D) FCFS

Q.25 [2 Marks] [MODERATE] [CN — Network Security]

Which data structure implements FIFO?

A) Stack
B) Queue
C) Tree
D) Graph

Q.26 [2 Marks] [MODERATE] [TOC — Grammars]

Time complexity of binary search?

A) O(n)
B) O(log n)
C) O(1)
D) $O(n^2)$

Q.27 [2 Marks] [MODERATE] [COA — Pipelining]

Normal form where all non-prime attributes depend on key:

A) 1NF
B) 2NF
C) 3NF
D) BCNF

Q.28 [2 Marks] [MODERATE] [Digital Logic — Combinational Circuits]

In deadlock prevention, hold-and-wait is prevented by:

A) Preemption

B) Resource ordering

C) Mutual exclusion

D) Allocation at once

Q.29 [2 Marks] [MODERATE] [Engineering Mathematics — Probability]

Class A IP range:

A) 0-127
B) 128-191
C) 192-223
D) 224-239

Q.30 [2 Marks] [MODERATE] [Programming — OOP]

A regular grammar generates:

A) Type 0
B) Type 1
C) Type 2
D) Type 3

Q.31 [2 Marks] [MODERATE] [Algorithms — Divide & Conquer]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

A) 0.1
B) 0.5
C) 0.9
D) 0.99

Q.32 [2 Marks] [MODERATE] [Data Structures — Linked Lists]

Which is not a page replacement algorithm?

A) FIFO
B) LRU
C) Optimal
D) FCFS

Q.33 [2 Marks] [HARD] [DBMS — SQL]

Which data structure implements FIFO?

A) Stack
B) Queue
C) Tree
D) Graph

Q.34 [2 Marks] [HARD] [OS — Process Sync]

Time complexity of binary search?

A) O(n)
B) O(log n)
C) O(1)
D) O(n^2)

Q.35 [2 Marks] [HARD] [CN — Network Security]

Normal form where all non-prime attributes depend on key:

A) 1NF

B) 2NF

C) 3NF

D) BCNF

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [TOC — Pushdown Automata]

Which are NP-Complete problems?

A) Hamiltonian Cycle
B) Sorting
C) Clique
D) Binary Search
E) Boolean SAT

Q.37 [2 Marks] [HARD] [COA — Pipelining]

Which are stable sorting algorithms?

A) Bubble
B) Insertion
C) Merge
D) Quick
E) Selection

Q.38 [2 Marks] [HARD] [Digital Logic — Sequential Circuits]

Which are valid lossless join decompositions?

A) BCNF
B) 3NF
C) 2NF
D) 1NF
E) All always lossless

Q.39 [2 Marks] [HARD] [Engineering Mathematics — Numerical Methods]

TCP provides:

A) Connection-oriented
B) Reliable
C) Unreliable
D) Flow control
E) Error recovery

Q.40 [2 Marks] [HARD] [Programming — Data Structures]

Which are NP-Complete problems?

A) Hamiltonian Cycle
B) Sorting
C) Clique
D) Binary Search
E) Boolean SAT

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Algorithms — Time Complexity]

Array A[1..100] traversed once. Time complexity?

Answer: ____________________

Q.52 [2 Marks] [HARD] [Data Structures — Graphs]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: ____________________

Q.53 [2 Marks] [HARD] [DBMS — SQL]

GATE CSE 2023: Semaphore with S=3, 5 processes need 1 each. How many wait?

Answer: ____________________

Q.54 [2 Marks] [HARD] [OS — Memory Management]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: ____________________

Q.55 [2 Marks] [HARD] [CN — HTTP/HTTPS]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: ____________________

Q.56 [2 Marks] [HARD] [TOC — Regular Languages]

Array A[1..100] traversed once. Time complexity?

Answer: ____________________

Q.57 [2 Marks] [HARD] [COA — Instruction Formats]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: ____________________

Q.58 [2 Marks] [HARD] [Digital Logic — Combinational Circuits]

GATE CSE 2023: Semaphore with S=3, 5 processes need 1 each. How many wait?

Answer: ___________________________

Q.59 [2 Marks] [HARD] [Engineering Mathematics — Linear Algebra]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: ______________________

Q.60 [2 Marks] [HARD] [Programming — C Programming]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: ______________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	**Q.2:**	B	**Q.3:**	B
Q.4:	D	**Q.5:**	A	**Q.6:**	D
Q.7:	C	**Q.8:**	D	**Q.9:**	B
Q.10:	B	**Q.11:**	B	**Q.12:**	D
Q.13:	A	**Q.14:**	D	**Q.15:**	C
Q.16:	D	**Q.17:**	B	**Q.18:**	B
Q.19:	B	**Q.20:**	D	**Q.21:**	A
Q.22:	D	**Q.23:**	C	**Q.24:**	D
Q.25:	B	**Q.26:**	B	**Q.27:**	B
Q.28:	D	**Q.29:**	A	**Q.30:**	D
Q.31:	C	**Q.32:**	D	**Q.33:**	B
Q.34:	B	**Q.35:**	B	**Q.36:**	ACE
Q.37:	ABC	**Q.38:**	ABC	**Q.39:**	ABDE
Q.40:	ACE	**Q.41:**	ABC	**Q.42:**	ABC
Q.43:	ABDE	**Q.44:**	ACE	**Q.45:**	ABC
Q.46:	ABC	**Q.47:**	ABDE	**Q.48:**	ACE
Q.49:	ABC	**Q.50:**	ABC	**Q.51:**	100
Q.52:	7	**Q.53:**	2	**Q.54:**	5
Q.55:	9	**Q.56:**	100	**Q.57:**	7
Q.58:	2	**Q.59:**	5	**Q.60:**	9

Detailed Solutions

Question 1 *[Algorithms — Divide & Conquer]*

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 2 *[Data Structures — Hashing]*

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' O(log n).

Question 3 *[DBMS — Indexing]*

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 4 *[OS — Deadlock]*

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 5 *[CN — Routing]*

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 6 *[TOC — Decidability]*

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 7 *[COA — Pipelining]*

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate × Miss penalty = 1 + (1-h)×100 = 10 !' h = 0.9.

Question 8 *[Digital Logic — Combinational Circuits]*

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 9 *[Engineering Mathematics — Discrete Math]*

Which data structure implements FIFO?

Answer:

(B)

Queue follows First-In-First-Out (FIFO) principle.

Question 10 *[Programming — C Programming]*

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' O(log n).

Question 11 *[Algorithms — Greedy]*

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 12 *[Data Structures — Linked Lists]*

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 13 *[DBMS — Concurrency]*

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 14 *[OS — Process Sync]*

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 15 *[CN — TCP/IP]*

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate × Miss penalty = 1 + (1-h)×100 = 10 !' h = 0.9.

Question 16 *[TOC — Grammars]*

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 17 *[COA — Addressing Modes]*

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 18 *[Digital Logic — Minimization]*

Time complexity of binary search?

Answer:

(B)

Binary search halves search space each step !' O(log n).

Question 19 *[Engineering Mathematics — Probability]*

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 20 *[Programming — Recursion]*

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 21 *[Algorithms — Dynamic Programming]*

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 22 *[Data Structures — Graphs]*

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 23 *[DBMS — Normalization]*

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate × Miss penalty = 1 + (1-h)×100 = 10 !' h = 0.9.

Question 24 *[OS — Deadlock]*

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 25 *[CN — Network Security]*

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 26 *[TOC — Grammars]*

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' O(log n).

Question 27 *[COA — Pipelining]*

Normal form where all non-prime attributes depend on key:

Answer:

(B)

2NF: no partial dependency of non-prime attributes on key.

Question 28 *[Digital Logic — Combinational Circuits]*

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 29 *[Engineering Mathematics — Probability]*

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 30 *[Programming — OOP]*

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 31 *[Algorithms — Divide & Conquer]*

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate × Miss penalty = 1 + (1-h)×100 = 10 !' h = 0.9.

Question 32 *[Data Structures — Linked Lists]*

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 33 *[DBMS — SQL]*

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 34 *[OS — Process Sync]*

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' O(log n).

Question 35 *[CN — Network Security]*

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 36 *[TOC — Pushdown Automata]*

Which are NP-Complete problems?

Answer:

(A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 37 *[COA — Pipelining]*

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 38 *[Digital Logic — Sequential Circuits]*

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 39 *[Engineering Mathematics — Numerical Methods]*

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 40 *[Programming — Data Structures]*

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 41 *[Algorithms — Dynamic Programming]*

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 42 *[Data Structures — Graphs]*

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 43 *[DBMS — Transactions]*

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 44 *[OS — Memory Management]*

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 45 *[CN — Routing]*

Which are stable sorting algorithms?

Answer:

(A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 46 *[TOC — Decidability]*

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 47 *[COA — Instruction Formats]*

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 48 *[Digital Logic — Minimization]*

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 49 *[Engineering Mathematics — Linear Algebra]*

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 50 *[Programming — Recursion]*

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 51 *[Algorithms — Time Complexity]*

Array A[1..100] traversed once. Time complexity?

Answer: (100)

Traversing n elements !' O(n) = 100 operations.

Question 52 *[Data Structures — Graphs]*

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: (7)

Full binary tree at height 2 has 7 nodes (perfect binary tree).

Question 53 *[DBMS — SQL]*

GATE CSE 2023: Semaphore with S=3, 5 processes need 1 each. How many wait?

Answer: (2)

3 proceed immediately. 2 wait. Total blocked = 2.

Question 54 *[OS — Memory Management]*

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer:

(5)

20 bytes / 4 bytes per word = 5 words.

Question 55 *[CN — HTTP/HTTPS]*

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: (9)

Partitioning n elements requires n-1 comparisons. For 10: 9.

Question 56 *[TOC — Regular Languages]*

Array A[1..100] traversed once. Time complexity?

Answer: (100)

Traversing n elements !' O(n) = 100 operations.

Question 57 *[COA — Instruction Formats]*

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: (7)

Full binary tree at height 2 has 7 nodes (perfect binary tree).

Question 58 *[Digital Logic — Combinational Circuits]*

GATE CSE 2023: Semaphore with S=3, 5 processes need 1 each. How many wait?

Answer: (2)

3 proceed immediately. 2 wait. Total blocked = 2.

Question 59 *[Engineering Mathematics — Linear Algebra]*

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: (5)

20 bytes / 4 bytes per word = 5 words.

Question 60 *[Programming — C Programming]*

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: (9)

Partitioning n elements requires n-1 comparisons. For 10: 9.
